

```

1 --s23_3b behavioral implementation
2
3 LIBRARY ieee;
4 USE ieee.std_logic_1164.ALL;
5 USE ieee.numeric_std.ALL;
6
7 ENTITY s23_3b IS
8     PORT (
9         clk : IN STD_LOGIC;
10        u : IN STD_LOGIC;
11        x : IN STD_LOGIC_VECTOR(2 DOWNTO 0);
12        y : OUT STD_LOGIC_VECTOR(2 DOWNTO 0)
13    );
14END ENTITY;
15
16ARCHITECTURE behavioral OF s23_3b IS
17    SIGNAL q : STD_LOGIC_VECTOR(2 DOWNTO 0) := (OTHERS => '0');
18BEGIN
19
20    -- sequential process
21    seq_p : PROCESS (clk)
22    BEGIN
23        IF rising_edge(clk) THEN
24            IF u = '1' THEN
25                q <= x;
26            ELSE
27                q(0) <= q(0);
28                q(1) <= q(1) XOR q(0);
29                q(2) <= q(2) XOR (q(1) OR q(0));
30            END IF;
31        END IF;
32    END PROCESS;
33
34    --combinational process
35    y <= q;
36
37END ARCHITECTURE behavioral;

```